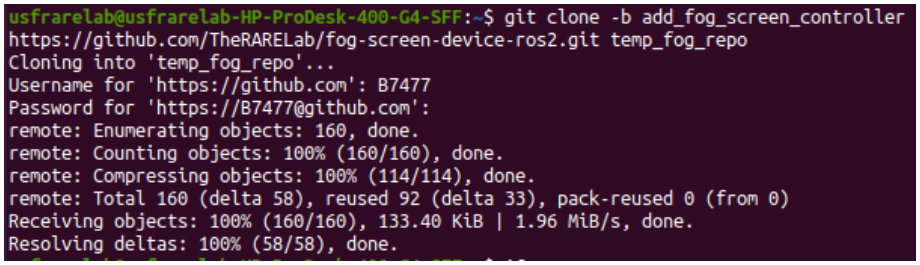
1. https://docs.ros.org/en/foxy/Installation/Ubuntu-Install-Debians.html
    a. Install ros2 packages and environment setup
2. Install ros2 foxy and dependencies
    a. sudo apt update
    b. sudo apt install python3-pip
    c. pip3 install pyserial
3. Create ros2 workspace (if does not exist already)
    a. https://docs.ros.org/en/foxy/Tutorials/Beginner-Client-Libraries/Creating-A-Workspace/Creating-A-Workspace.html
4. Set up ROS2 workspace
    a. mkdir -p ~/ros2_ws/src
        i. Always put any packages in your workspace into the src directory
    b. cd ~/ros2_ws/src
5. Cloning repo from GitHub onto workspace (TWO OPTIONS)
    a. **Option A**: Quick clone (Warning: extra files will also be cloned)
        i. Ensure you're still in the ros2_ws/src directory before you clone.
        ii. git clone https://github.com/TheRARELab/fog-screen-device-ros2.git -b add_fog_screen_controller
            1. Will clutter your workspace and possible cause ROS2 package build errors if this method is used
    b. **Option B (Recommended):** Clean clone using sparse checkout or manual copy
        i. Cd ~
        ii. git clone -b add_fog_screen_controller https://github.com/TheRARELab/fog-screen-device-ros2.git temp_fog_repo
            1. 

            (PASSWORD IS A TOKEN CREATED BY USER, NOT USERS PASSWORD)
        iii. cp -r ~/temp_fog_repo/fog_screen_controller ~/ros2_ws/src/

1.
```
usfrarelab@usfrarelab-HP-ProDesk-400-G4-SFF:~$ cd ros2_ws/src/
usfrarelab@usfrarelab-HP-ProDesk-400-G4-SFF:~/ros2_ws/src$ ls
fog_screen_controller
usfrarelab@usfrarelab-HP-ProDesk-400-G4-SFF:~/ros2_ws/src$ cd fog_screen_controller/
usfrarelab@usfrarelab-HP-ProDesk-400-G4-SFF:~/ros2_ws/src/fog_screen_controller$ ls
fog_screen_controller  package.xml.save  setup.cfg  test
package.xml            resource          setup.py
```
(just to show it has copied)

iv. rm -rf ~/temp_fog_repo
    1. Removes the temp folder (cleans up workspace)

6. Resolve dependencies - best practice is to check for dependencies every time you clone
    a. cd ~/ros2_ws
    b. rosdep install -i --from-path src --rosdistro foxy –y
        i. If already installed: #All required rosdeps installed successfully
            1. Dependencies declared in package.xml file; This command walks through those declarations and installs the ones that are missing.
7. Build package
    a. cd ~/ros2_ws
    b. colcon build --packages-select fog_screen_controller
        i.
```
usfrarelab@usfrarelab-HP-ProDesk-400-G4-SFF:~/ros2_ws$ colcon build --packages-select
fog_screen_controller
Starting >>> fog_screen_controller
Finished <<< fog_screen_controller [1.04s]

Summary: 1 package finished [1.42s]
```
(builds the package; similar to catkin_make in ROS1)

    c. source install/setup.bash
        i.
```
usfrarelab@usfrarelab-HP-ProDesk-400-G4-SFF:~/ros2_ws$ source install/setup.bash
ROS_DISTRO was set to 'foxy' before. Please make sure that the environment does not mi
x paths from different distributions.
ROS_DISTRO was set to 'noetic' before. Please make sure that the environment does not
mix paths from different distributions.
```
(so terminal knows where to find newly built packages)

        ii. So that you dont have to do this every time:
            1.
```
usfrarelab@usfrarelab-HP-ProDesk-400-G4-SFF:~/ros2_ws$ source ~/ros2_ws/install/
setup.bash
usfrarelab@usfrarelab-HP-ProDesk-400-G4-SFF:~/ros2_ws$ source ~/.bashrc
```
    d. source /opt/ros/foxy/setup.bash
        i.
```
usfrarelab@usfrarelab-HP-ProDesk-400-G4-SFF:~/ros2_ws$ source /opt/ros/foxy/setup.ba
sh
ROS_DISTRO was set to 'noetic' before. Please make sure that the environment does no
t mix paths from different distributions.
```
8.
        i. Lab computer had noetic incharge before
9. Connect USB and in ~ (home) directory and type ls /dev/tty* (ls /dev/ttyACM0)
10. Power on fog machine, plug in DMX cable
11. sudo usermod -a -G dialout $USER AND THEN REBOOT (sudo reboot)

12.
```
usfrarelab@usfrarelab-HP-ProDesk-400-G4-SFF:~/ros2_ws$ ros2 run fog_screen_controller fog_node
[INFO] [1756499757.469383778] [fog_device_controller]: Fog machine turned OFF
[INFO] [1756499832.007768900] [fog_device_controller]: Fog machine turned ON
[INFO] [1756499835.077074501] [fog_device_controller]: Fog machine turned OFF
```

In main terminal

13. Terminal where service is called

```
usfrarelab@usfrarelab-HP-ProDesk-400-G4-SFF:~/ros2_ws$ ros2 service list
/fog_device_controller/describe_parameters
/fog_device_controller/get_parameter_types
/fog_device_controller/get_parameters
/fog_device_controller/list_parameters
/fog_device_controller/set_parameters
/fog_device_controller/set_parameters_atomically
/fog_machine/turn_off
/fog_machine/turn_on
usfrarelab@usfrarelab-HP-ProDesk-400-G4-SFF:~/ros2_ws$ ros2 node list
/fog_device_controller
usfrarelab@usfrarelab-HP-ProDesk-400-G4-SFF:~/ros2_ws$ ros2 service call /fog_machine/turn_off
 std_srvs/srv/Empty
waiting for service to become available...
requester: making request: std_srvs.srv.Empty_Request()

response:
std_srvs.srv.Empty_Response()

usfrarelab@usfrarelab-HP-ProDesk-400-G4-SFF:~/ros2_ws$ ros2 service call /fog_machine/turn_on
std_srvs/srv/Empty
waiting for service to become available...
requester: making request: std_srvs.srv.Empty_Request()

response:
std_srvs.srv.Empty_Response()

usfrarelab@usfrarelab-HP-ProDesk-400-G4-SFF:~/ros2_ws$ ros2 service call /fog_machine/turn_off
 std_srvs/srv/Empty
requester: making request: std_srvs.srv.Empty_Request()

response:
std_srvs.srv.Empty_Response()
```